Mullvad Split Tunneling and VM Bypass Notes

Mullvad split tunneling

Mullvad split tunneling can be configured through the GUI or through the Mullvad CLI.

Run Command Prompt or PowerShell as Administrator.

```
mullvad split-tunnel get
```

Shows whether split tunneling is enabled and lists excluded applications.

```
mullvad split-tunnel set on
mullvad split-tunnel set off
```

Turns split tunneling on or off.

```
mullvad split-tunnel app add "C:\Program Files\Mozilla Firefox\firefox.exe"
```

Adds an application to the excluded apps list.

```
mullvad split-tunnel app remove "C:\Program Files\Mozilla Firefox\firefox.exe"
```

Removes an application from the excluded apps list.

```
mullvad split-tunnel app clear
```

Clears all excluded applications.

Excluded applications use the normal internet connection instead of the VPN tunnel, so they expose the normal public IP address.

Mullvad config file

Some Mullvad GUI settings may be visible here:

```
%LOCALAPPDATA%\Mullvad VPN\gui_settings.json
```

Do not rely on editing this file manually. Use the Mullvad GUI or CLI instead, because manual changes may be overwritten or may not properly update the Mullvad service.

Important Mullvad settings

Lockdown mode

Lockdown mode should be off if you want non-VPN traffic to work.

If lockdown mode is on, Mullvad can block internet access that is not going through the VPN.

Local network sharing

Local network sharing should usually be on if you want access to local network devices such as:

```
192.168.x.x devices
routers
printers
local VMs
NAS devices
```

Excluding virtual machines from Mullvad

Normal Mullvad split tunneling works by excluding programs. A whole VM is different because the VM uses virtual network adapters instead of behaving like one normal Windows application.

For VMs, the cleaner method is to give the VM a bridged/external network path through a physical network adapter.

Hyper-V external switch for VM bypass

Create an external Hyper-V switch bound to the physical adapter:

```
New-VMSwitch -Name "VM-Bypass-Mullvad" -NetAdapterName "Ethernet" -
AllowManagementOS $true
```

Check existing switches:

```
Get-VMSwitch | Format-Table Name,SwitchType,NetAdapterInterfaceDescription,Id
```

Use the switch ID and switch name in the VM configuration.

NanaBox network config example

```
"NetworkAdapters": [
  {
    "Connected": true,
    "EndpointId": "",
    "MacAddress": "YOUR_MAC_ADDRESS",
    "SuggestedSwitchId": "YOUR_SWITCH_ID",
    "SuggestedSwitchName": "VM-Bypass-Mullvad"
  }
]
```

Notes:

```
EndpointId is generated when the VM runs.
MacAddress can be chosen manually.
SuggestedSwitchId should match the Hyper-V switch ID.
SuggestedSwitchName should match the Hyper-V switch name.
```

This should give the VM a normal local network IP address such as:

```
192.168.x.x
```

Hyper-V Manager method

1. Open Hyper-V Manager.
2. Open the VM settings.
3. Go to Network Adapter.
4. Set the virtual switch to `VM-Bypass-Mullvad`.
5. Start the VM and check that it receives a LAN IP address.

VirtualBox options

Bridged directly to Realtek

```
VM → VirtualBox bridge driver → Realtek physical adapter → router
```

This gives the VM direct LAN access through the physical adapter.

Bridged to Hyper-V vEthernet adapter

Use this command to map adapters:

```
Get-NetAdapter | Format-Table Name,InterfaceDescription,ifIndex,MacAddress
```

Path:

```
VM → VirtualBox bridge driver → Hyper-V vEthernet adapter → Hyper-V external
switch → Realtek adapter → router
```

Testing

Inside the VM, check the local IP:

```
ipconfig
```

The VM should have a normal LAN IP, usually like:

```
192.168.x.x
```

Then check whether the VM is using Mullvad or the normal connection by visiting Mullvad's connection check page from inside the VM.

Expected result for a bypassed VM:

```
Not using Mullvad VPN
```